Segmentation Annotations For DL (FAQ)

Where can/should I generate manual-segmentions?

WebKnossos

The basics…

WebKnossos is a web-based platform designed for visualizing, annotating, and sharing 3D imaging datasets. It allows users to annotate large electron-microscopy (EM) images in a web browser without local installation.

Here's a link to the 'Beginners Guide' - it will give you basic information on signing up for a free account, uploading datasets (50GB storage limit for free accounts) and use of basic annotation tools.

- https://docs.webknossos.org/webknossos/getting_started.html

Here is some good documentation from WebKnossos on how to make 'bounding boxes' used for custom model training…

- https://docs.webknossos.org/webknossos/automation/choosing_mags_and_bboxes.html

Additionally, here is some generally helpful documentation detailing the 'best-practices' for 'Annotating Ground Truth Data for ML in WEBKNOSSOS'.

- https://docs.webknossos.org/webknossos/tutorials/trainingdata_annotation.html
- I like this tutorial as they provide a nice explanation for what you should do to maintain consistency across annotations and reasons why we need training data to capture membrane/organelle diversity.
- Note, they want you to use their pay-to-play models but ignore that part and just focus on the annotation portion.

Exporting Annotations…

There are a few ways/formats in which we can export annotations (bounding boxes and their labels) from the Webknossos UI, including Zarr downloads, Tiff/OME-Tiff export, and a python API. However, the most relevant for us will be the Tiff/OME-Tiff export.

Thankfully, WebKnossos has pretty good documentation on this, so visit the link below for more information.

- https://docs.webknossos.org/webknossos/data/export_ui.html

Additionally, if you're interested in utilizing the 'ZAR streaming capabilities of Webknossos, read more about that at the link below.

https://docs.webknossos.org/webknossos/data/zarr_stream.html

Speeding up annotations…

WebKnossos features a built-in smart (AI assisted) annotation tool which can selectively annotate ultrastructures (nucleus, mitochondria, etc) and even individual cells using 'quick selection.

WebKnossos also features an interpolation tool, which allows users to interpolate the annotations between two different slices in the z-stack (e.g. - you can annotate z-slice 10 and z-slice 15, then interpolate the annotations between the two).

- Note, this method is not flawless, interpolations should be checked as quality control

Below is the documentation for using these features (both are in the same section of the documentation linked).

- Volume annotations - https://docs.webknossos.org/webknossos/tutorial_volume_annotation.html#smart-annotation-tools
- General annotation tools - https://docs.webknossos.org/webknossos/volume_annotation/tools.html

DragonFly

DragonFly is another one of these softwares for visualizing, annotating, and training deep learning segmentation models.

Here is a link to their landing page for 'life scientists', just for reference…

- https://dragonfly.comet.tech/en/applications/life-sciences

DragonFly does have a free version for 'non-commercial' academic purposes, but this licence is limited to a single machine (not compatible with a cluster/virtual-machine).
Here is the link to where you can submit a short application for a licence to download the free version.

- https://dragonfly.comet.tech/en/free-trial-dragonfly-3d-world

DragonFly appears to be the more 'hefty' approach to our segmentation problems.
While even the free version of DragonFly is a powerful tool, it requires a significant amount of computational resources to run. See the 'System Requirements/Recommendations' page below for more information.

- https://dragonfly.comet.tech/en/support-and-training/system-requirements
- Only supports Windows or Linux (no Mac?)
- You NEED at least 1GPU - but I would recommend more, each with a decent amount of RAM
- It's recommended to 32+ GB of cpu RAM (plan on having 4x as much RAM as the largest scans you need to process).
- 2 TB of local hard disk storage

What Makes 'Good' Annotations for a Deep-Learning Segmentation Model? [Draft 1 - please comment more questions and I will answer]

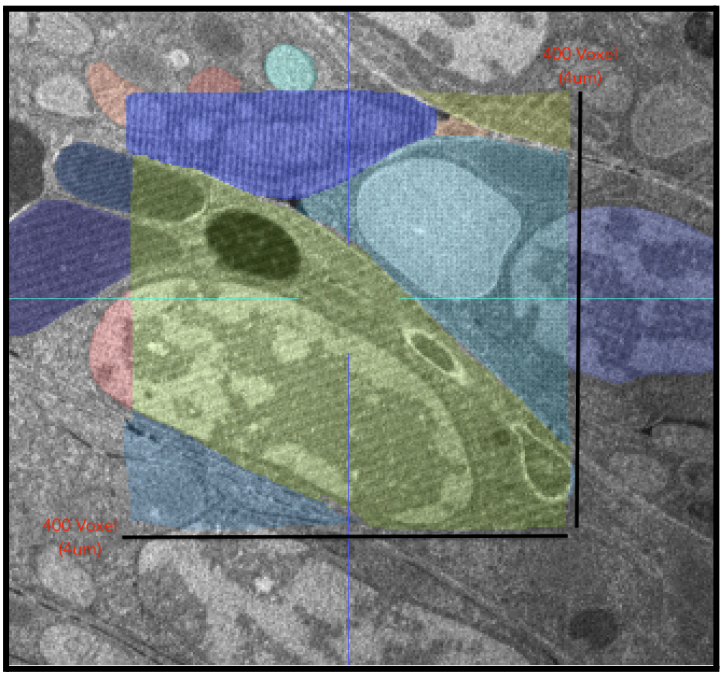
For the purposes of this FAQ/Tutorial we will refer to the annotations of the *Clytia* planula dataset provided by Shreya Rai from the Houliston/Momose lab at IMEV, obtained at the EMBL core facility in collaboration with Paolo Ronchi. This is unpublished so please do not share access to it beyond the ZooCell Network.

*Certain information below will be copied or rephrased from the WebKnossos documentation since they did a pretty good job

Dimensions

Our deep learning models only 'see' a very limited window of voxels during training/prediction. This window, also known as the kernel, slides across the entirety of our input data to encode it in a format for which its features are then learnable by our models.

Thus, we do not need to annotate the entirety of an XY plane across countless Z-slices. Instead, we can use densely annotated cubes which capture enough of the membrane/cell shape variation we need for training a model which generalizes well to the whole dataset.



For the purposes of Shreya's dataset, which has a voxel size of 10nm isotropic (10x10x10), we chose to make densely annotated cubes in the dimensions of 400 voxels^3 (400x400x400, isotropic)

The reason for this is…

1. We want a wide enough window to capture several membrane shapes/interactions without becoming a complete burden for those doing the annotations
    a. **More spatial context** typically improves model learning.
2. Densely annotated blocks are more efficient for training
3. We have similar sized blocks have been used before to train good segmentation models before

Quantity

- Quick answer - We recommend at least 3-4 densely annotated cubes of the dimensions mentioned above ^

Diversity

How does my choice of training data affect model performance?

- Quick answer - The more diversity in membrane shapes and interactions represented in training data, the better the model should generalize to the rest of the dataset.
- When selecting your blocks try to sample from different morphological regions throughout the volume to capture as much of that diversity as possible. See the three regions selected by Shreya as a reference, below…

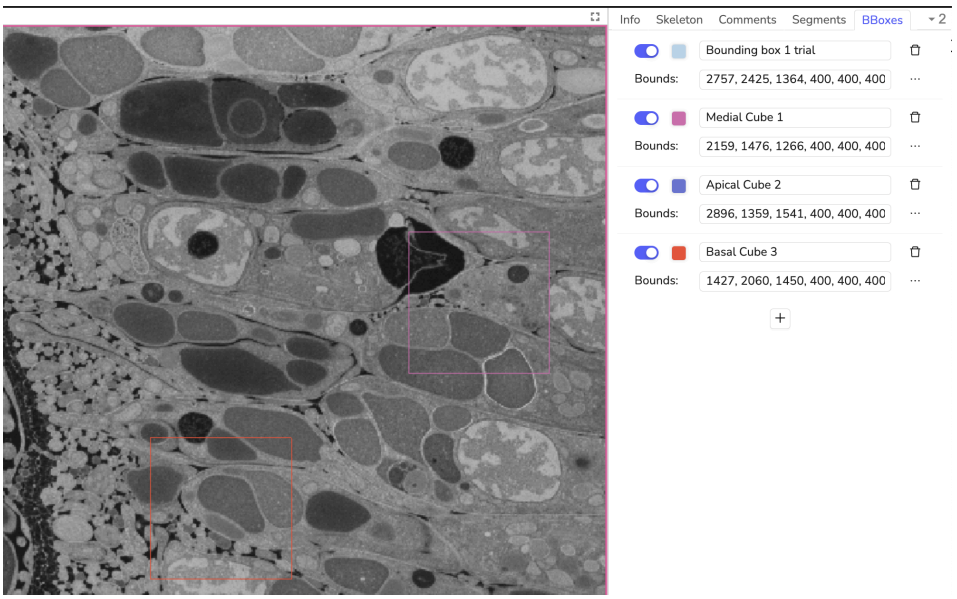
Do I need to consider the ultrastructure content of cells if I'm just annotating the whole cell?

- Quick answer - No. If you are just training a cell(-membrane) segmentation model then you don't need to worry about whether you have enough nuclei or other ultrastructures for other segmentation tasks.
    - A model trained from segmentations of the whole cell will only produce boundary probabilities for cell membranes.
    - The model simply predicts the likelihood of whether voxels belong to a membrane or not (probability between 0-1).
    - Likewise, if you train a model to segment chromatin, it will (hopefully) only segment chromatin.

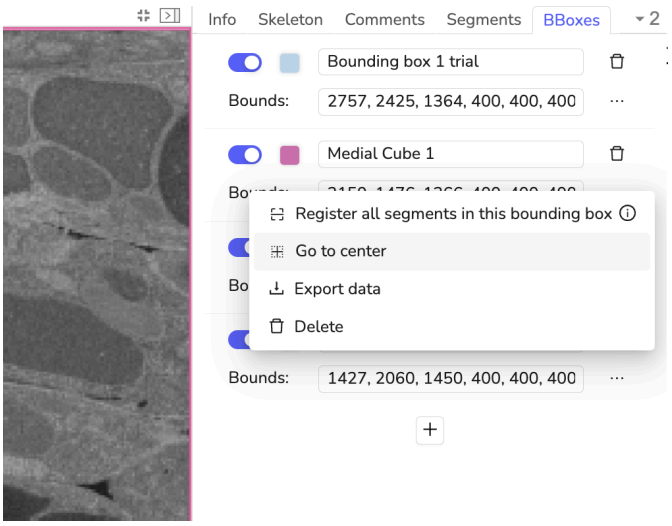
Examples Regions Selected by Shreya on the *Clytia* planula:

- Apical Cube - Apical_cube2.avi
- Medial Cube - Medial_cube1.avi
- Basal Cube - Basal_cube3.avi

You can view these cubes in webknossos to see their position within the full dataset:
https://webknossos.org/links/nuKr-LY5kYg5W4sW



Please select 'Bboxes' in the right sidebar.
Click the three-dot (ellipsis) menu to go to the centre of a specific cube.



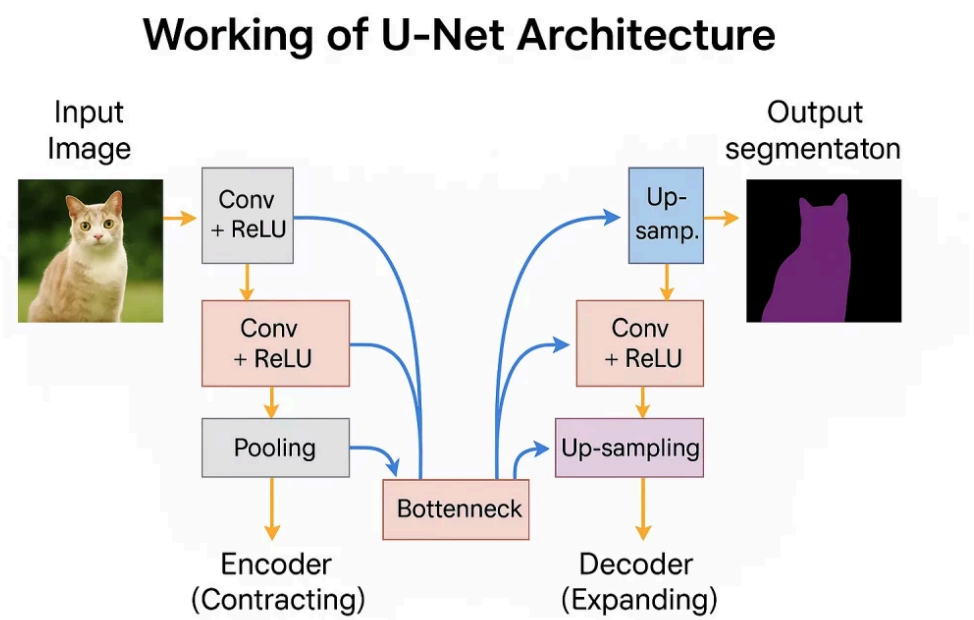
Accessibility

- Quick answer - wherever you want to annotate is fine as long as the volume and annotations are accessible and can be exported for downstream training tasks.
    - We are using WebKnossos for the *Clytia* dataset.
- Additionally, ensure that you can export the annotations and volumes in a format that is usable by the ilastik pipeline (for now).
    - This includes the file formats compressed-hdf5, hdf5, OME-zarr, and n5 (less preferred).

What Are the Segmentation Models and What Are They Actually Predicting? [Draft 1 - please comment more questions and I will answer]

What is the technical term for the 3D Segmentation models we use?

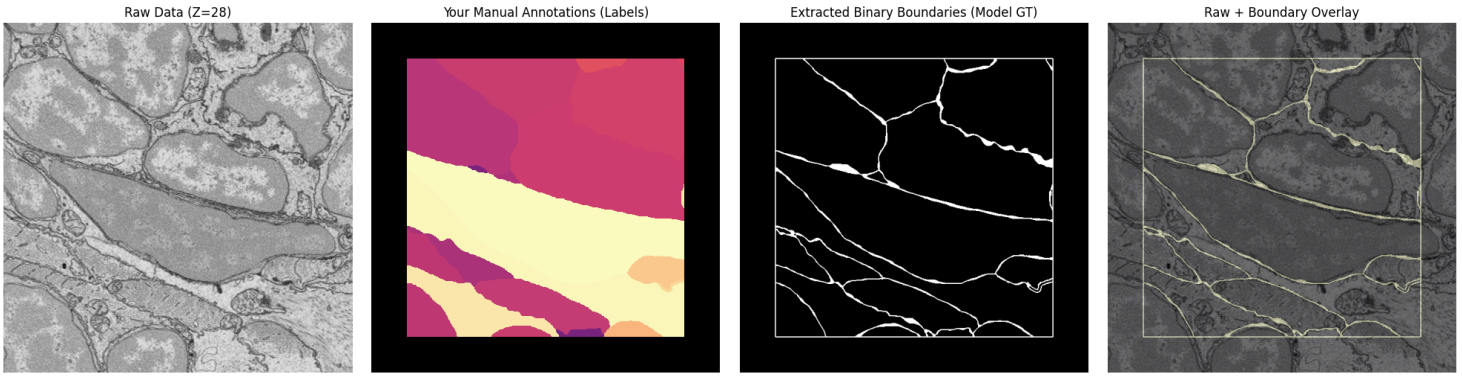
- Quick answer - there are many models which can be used for segmentation tasks, but we preferentially use 3D U-Nets (which is considered a deep learning neural network) to train segmentation models.
- Specifically, U-Nets are a type of convolutional neural network (CNN) designed for image segmentation, characterized by a U-shaped architecture



    - Here is an (ok) diagram to simply visualize the input/output and 'U' shape architecture of these networks.
- For more info…
    - see this wikipedia article for a layman summary: https://en.wikipedia.org/wiki/U-Net
    - or watch one these YouTube Videos:
        - https://www.google.com/url?sa=t&source=web&rct=j&opi=89978449&url=https://www.youtube.com/watch%3Fv%3DNhdzGfB1q74&ved=2ahUKEwj7tMv05bqTAxVbgf0HHThtLrMQtwJ6BAgYEAI&usg=AOvVaw2kdOh5lSMXBGzd4xseClB5 - supposed to simplier/shorter (10mins)
        - https://www.google.com/url?sa=t&source=web&rct=j&opi=89978449&url=https://www.youtube.com/watch%3Fv%3DNhdzGfB1q74&ved=2ahUKEwj7tMv05bqTAxVbgf0HHThtLrMQtwJ6BAgYEAI&usg=AOvVaw2kdOh5lSMXBGzd4xseClB5 - longer, more complex and w/ application examples (30mins)

What do these segmentation models predict?

- Quick answer - as noted earlier, the model predicts the likelihood of whether voxels belong to a boundary (membrane, organelle, or whatever other boundary it's trained to predict) or not, with a probability between 0-1.



A B C D

A. This is an example of a z-slice from the ground-truth cubes used to train the *platyneris* segmentation model.
B. This is an example of dense labeling for the same slice in that cube.
C. This is an example of the binary ground truth boundaries that we can extract from the densely labeled cube. Note, the white represents the presence of a boundary (1) and the black is the background/absence of a boundary (0).
D. This is a simple overlay of the binary ground truth over the raw data, just so you can see how the boundaries map on to our original image.

How do I get from boundary probabilities to my final prediction? [In-Progress]

- Quick answer - there are a number of necessary post-processing steps required to go from the extracted boundary probabilities to a full labeled/segmented volume - including Watershed and Multicut
    - We will discuss this in much more detail during the Naples course.
    - If you are interested in reading more about this now, try checking out the documentation for the multicut workflow at their website. It's probably the most succinct example you'll find on the internet without using an LLM to break it down for you - https://www.ilastik.org/documentation/multicut/multicut